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AI for Bharat Hackathon

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Team Name : Hack/Slash

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Problem Statement : Gram Sathi - AI for Rural Innovation & Sustainable Systems

Gram Sathi - AI for Millions of Indians Who Can't Access AI

PROBLEM

- 150M+ Indian farmers have no smartphone, no internet, no access to AI — they rely on middlemen and guesswork for pricing, weather, and crop decisions.
- Every Agri-tech solution today assumes connectivity — apps, chatbots, and portals exclude the farmers who need them most.
- The result: farmers sell below market rate, miss weather warnings, and never learn about government schemes they qualify for.

INSIGHT

- Every one of them has a phone number
- India processes 5 billion missed calls daily
- it's the most instinctive digital action.

SOLUTION

One missed call, Free of cost, AI calls you back

- Farmer speaks in their language → AI agent fetches live mandi prices, weather
- Government scheme eligibility → Responds via voice/SMS. No app. No internet
- Proactive Alerts -> AI Calls/SMS Farmer weather warnings before storms

No smartphone. Just your voice.

Why Everything Else Fails — And Why We Don't

PRESENT TODAY

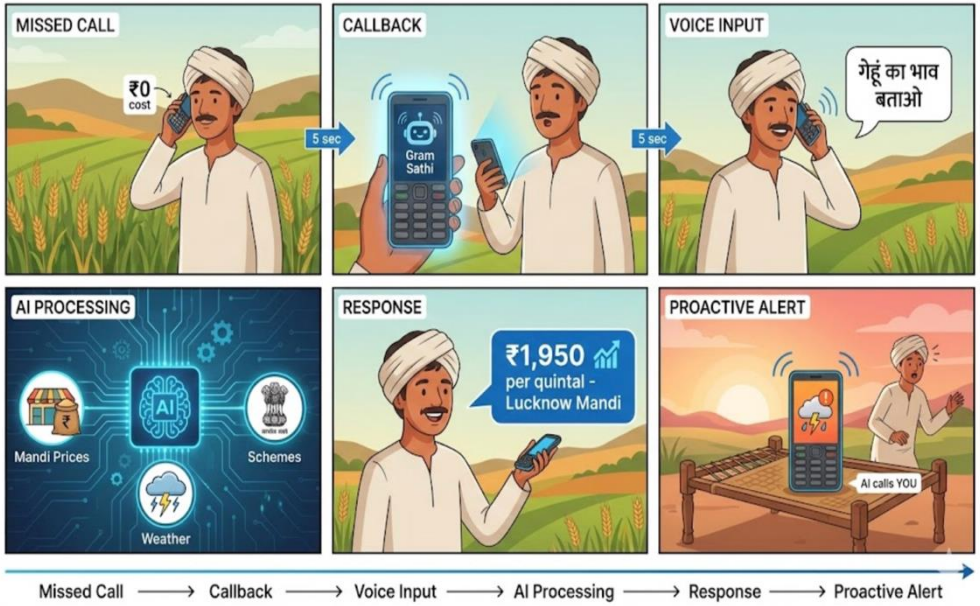
Farmer.Chat, KisanGPT, AgroTalk, Bharat-VISTAAR →

All need a smartphone
All need internet

VS

Gram Sathi →

Works on a ₹500 Nokia. Zero cost. Zero internet. Zero literacy needed.



HOW IT SOLVES THE PROBLEM:

Step 1: Farmer dials our number, hangs up **(₹0 — missed call)**

Step 2: AI detects missed call, calls back in 5 seconds **(we pay)**

Step 3: AI agent speaks in farmer's language **(22 languages, auto-detected)**

Step 4: Agent pulls LIVE data - mandi prices, weather, scheme eligibility

Step 5: Farmer gets actionable answer. Profile saved. Next call is smarter.

Features

🌾 LIVE MANDI PRICES

Real-time prices from 1,266 mandis via eNAM/data.gov.in.

Best mandi recommendation. Price trend alerts.

🌦 WEATHER INTELLIGENCE

5-day hyperlocal forecast. Rainfall, temp, humidity.

Severe weather warnings (hailstorm, frost, heatwave).

📋 SCHEME MATCHER (Amazon Q)

Auto-matches farmer profile against 1,000+ govt schemes.

PM-KISAN, PMFBY, Ayushman Bharat + state schemes.

🌱 CROP ADVISORY

Season & weather-aware sowing, irrigation, pest advice.

Region-specific recommendations.

🧠 SMART FARMER PROFILE

Remembers state, district, crops, land size across calls.

Subsequent calls after 1st call is instantly personalized. No re-explaining.

🔔 PROACTIVE ALERTS

AI CALLS/SMS THE FARMER — weather warnings before storms,

price spikes, scheme deadlines. Without being asked.

🗣 22-LANGUAGE VOICE AI

Auto-detects language. No menus. No 'Press 1 for Hindi'.

Sarvam AI + Bhashini. Voice-first, zero-UI.

📊 MONITORING DASHBOARD

Real-time call logs, user analytics, query insights.

For NGOs, govt officials, FPOs to track adoption.

Process Flow — From Missed Call to AI Response

USER JOURNEY (6 Steps):

1. MISSED CALL → Farmer dials Gram Sathi number, hangs up (₹0 cost)

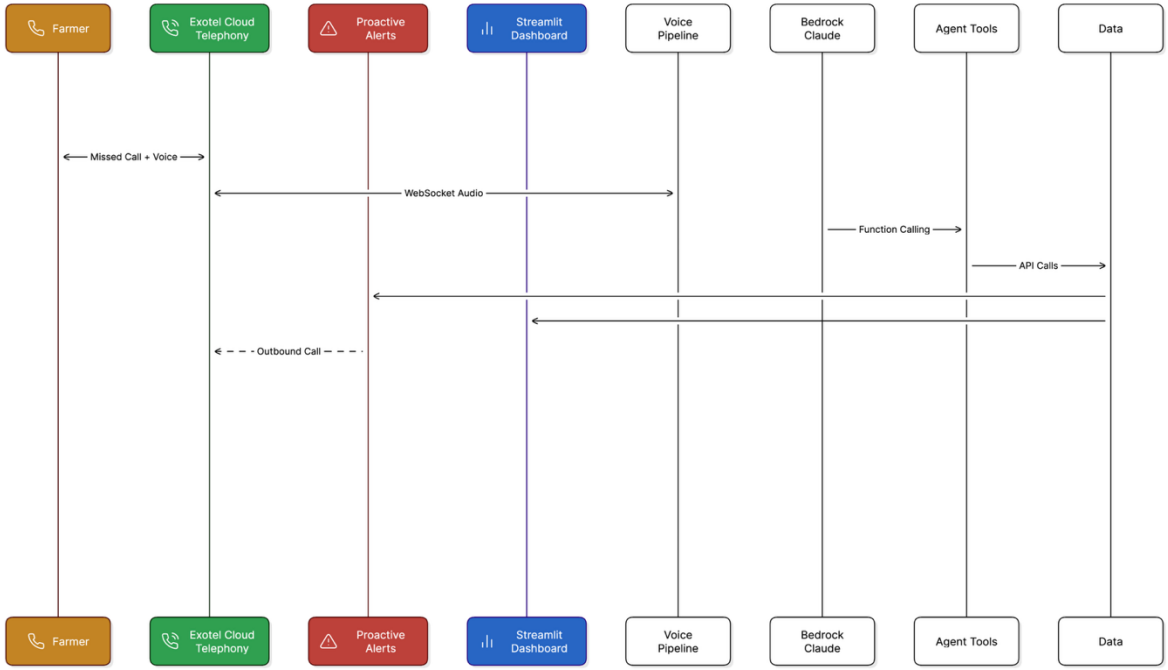
2. WEBHOOK → VOIP detects missed call, fires POST to our server

3. CALLBACK → Server calls farmer back in 5 seconds via Exotel outbound API

4. VOICE INPUT → Farmer speaks naturally in any language (Sarvam ASR auto-detects)

5. AI PROCESSING → Bedrock Claude processes intent, calls tools (prices/weather/schemes)

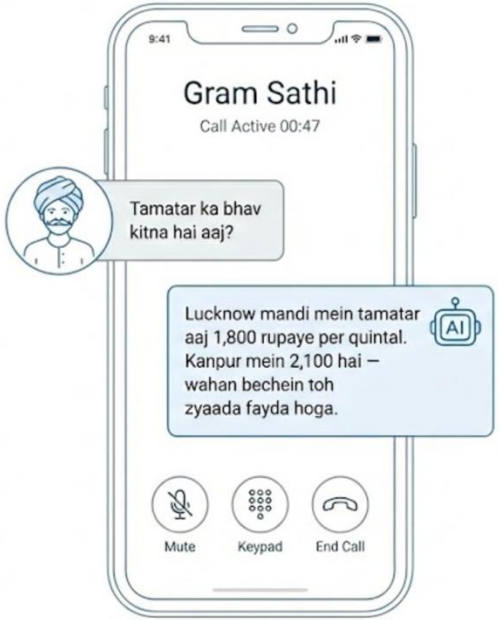
6. VOICE RESPONSE → AI responds with real data in farmer's language (Sarvam TTS)



Gram Sathi has TWO interfaces

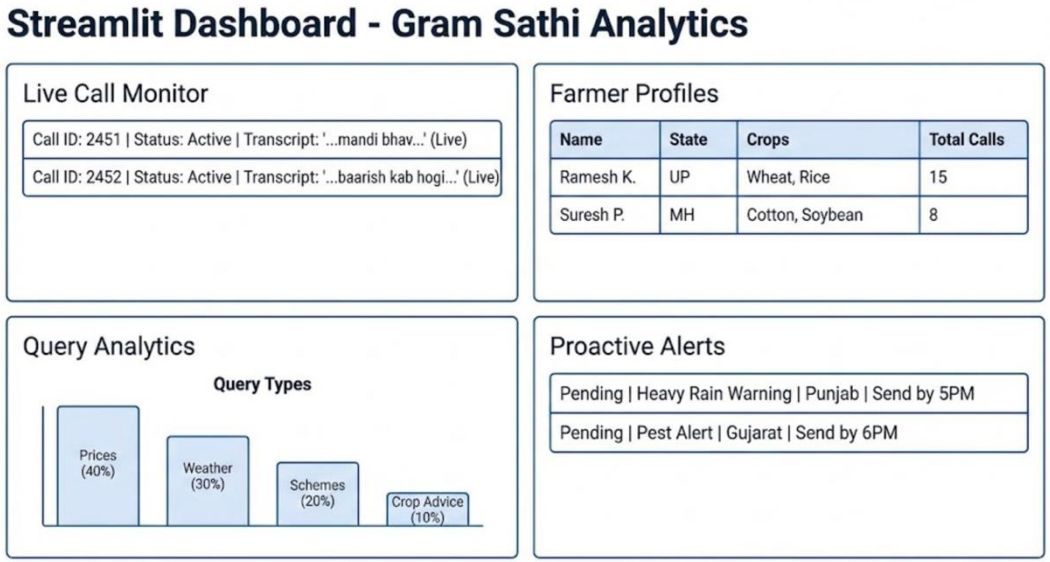
1. FARMER INTERFACE = The Phone Call (Zero UI)

No screens. No buttons. No app. The voice conversation IS the interface

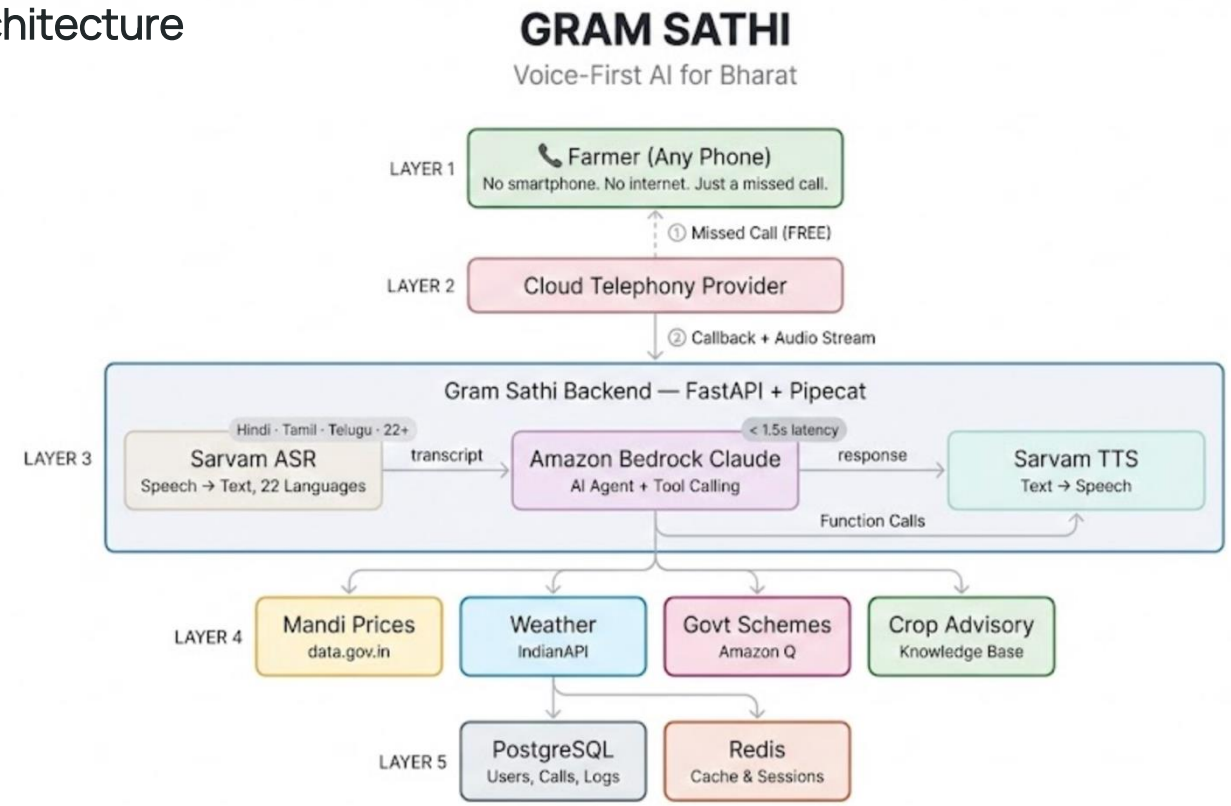


2. ADMIN DASHBOARD = Streamlit Web App (for NGOs/Govt)

Call logs, farmer profiles, query analytics, alert management.



System Architecture



Technology Stack

AWS SERVICES

Amazon Bedrock - Claude LLM (ap-south-1 Mumbai)
Function calling / tool use for live data retrieval

Amazon Q Business - Government scheme knowledge base
1,000+ schemes from MyScheme.gov.in ingested

Kiro IDE - Spec-driven development

TELEPHONY

Exotel - Indian cloud telephony (₹1000 free trial)
Missed call detection, outbound callbacks, WebSocket audio

VOICE AI

Sarvam AI Saaras v3 - ASR, 22 languages, < 1s latency

Sarvam AI Bulbul v3 - TTS, 11 languages, 30+ voices

Bhashini - Government language platform (fallback)

BACKEND

Python + FastAPI - Async, WebSocket support

Pipecat - Open-source voice pipeline framework

PostgreSQL - User profiles, call logs

Redis - Caching, session state, rate limiting

DATA SOURCES

data.gov.in API - Daily mandi prices (3000+ markets)

IndianAPI.in - Weather forecasts (free)

MyScheme.gov.in - HuggingFace dataset (1000+ schemes)

DASHBOARD

Streamlit - Call logs, analytics, user profiles

DEPLOYMENT

Docker + AWS EC2 (ap-south-1) - Low latency in India

Estimated Implementation Cost

Component	Assumption	Cost/Day (₹)	Cost/Month (₹)
Exotel Telephony	100 calls × 2 min × ₹2/min	400	~12,000
Sarvam AI ASR	3.3 hrs/day × ₹30/hr	100	~3,000
Sarvam AI TTS	~100K chars/day	50	~1,500
Amazon Bedrock (Claude)	200 inferences/day	100	~3,000
AWS EC2 (t3.medium, ap-south-1)	Standard instance cost	100	~3,000
PostgreSQL + Redis	RDS + ElastiCache	50	~1,500
data.gov.in, IndianAPI, Bhashini	Open APIs	FREE	FREE
TOTAL (MVP)		~800	~24,000

HACKATHON FREE TIER:

Exotel trial: ₹1,000 free | Sarvam AI: ₹1,000 free | AWS hackathon credits: TBD

Total hackathon cost: Effectively ₹0 with free tiers

Cost / Farmer Interaction: ~₹8 per call (2 min avg)

Value / Farmer Interaction : ₹100-1000+ (better prices, scheme access)

Impact & Scalability

IMMEDIATE IMPACT:

- Non smartphone users can access AI
- Farmers save 15-20% income lost to price information asymmetry
- Unclaimed govt schemes become discoverable
- Proactive weather alerts prevent crop losses worth thousands per farmer

SCALABILITY:

- Zero marginal distribution cost — no app store, no downloads
- One phone number scales to all of India (Exotel handles routing)
- Sarvam AI + Bhashini cover 22 Indian languages — same system
- Farmer profiles improve with every call — network effect on intelligence
- Government integration ready: AgriStack, ONDC, eNAM, PM-KISAN

THE VISION

'Every farmer in India, one missed call away from AI.'

If we reach 1% of rural India = 3 million farmers.

At ₹8/call, 3M farmers × 30 calls/year = ₹72 crore annual cost.

Value created: ₹3,000+ per farmer per year = ₹900+ crore value.

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Thank You